CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Topic 2.1: Types and Methods of Data Transmission • Practice Worksheet

Total Marks: 30 Marks • Time Allowed: 45 Minutes

Candidate Name: _____ **Date:** _____

Class / Section: _____ **Score:** _____ **/ 30 Marks**

Question 1: Data Packets and Packet Structure [6 Marks]

Data transmitted across the Internet is divided into packets.

(a) State the **three** primary parts that make up a data packet. [3]

1. _____
2. _____
3. _____

(b) Identify **three** items of information found in a packet header. [3]

1. _____
2. _____
3. _____

Question 2: Packet Switching Mechanism [4 Marks]

Describe how **packet switching** is used to transmit a large file across the Internet from a sender to a receiver. Include details on routing and reassembly.

_____

_____

_____

_____

Question 3: Direction of Data Transmission [6 Marks]

Data transmission can be **simplex**, **half-duplex**, or **full-duplex**.

Complete the table below by stating the correct transmission mode for each scenario and giving a reason:

Scenario	Transmission Mode	Reason
A student speaks to another student using a pair of traditional Walkie-Talkies.	_____	_____
A computer sends a print command and document data to a laser printer.	_____	_____
Two people talking to each other simultaneously during a live video call.	_____	_____

Question 4: Serial vs. Parallel Data Transmission [6 Marks]

(a) Tick (✓) one box in each row to indicate whether the statement applies to **Serial** or **Parallel** transmission: [4]

Statement	Serial	Parallel
Bits are sent one bit at a time down a single wire or channel.	[ ]	[ ]
More suitable and reliable for long-distance communication.	[ ]	[ ]
Higher risk of bit skew occurring during transmission.	[ ]	[ ]
Commonly used for internal high-speed computer buses over very short distances.	[ ]	[ ]

(b) Explain what is meant by the term **bit skew** and why it causes problems in parallel transmission. [2]

Question 5: Universal Serial Bus (USB) [4 Marks]

Universal Serial Bus (USB) is an industry-standard interface used to connect peripheral devices to computers.

(a) State **two** benefits of using USB connections compared to older legacy ports. [2]

(b) State **two** drawbacks or limitations of using USB. [2]

Question 6: Transmission Application Scenario [4 Marks]

An underground scientific sensor station located 8 kilometers away from a university laboratory needs to transmit temperature data continuously to the laboratory main server.

(a) Recommend whether **serial** or **parallel** transmission should be used. Give **two** reasons for your recommendation. [3]

Recommendation: _____
Reason 1: _____
Reason 2: _____

(b) The laboratory needs to send software configuration updates back to the sensor station while receiving sensor readings at the same time. Name the direction mode required. [1]

Direction Mode: _____